

Automated Rubric Assignment Evaluator

Work Breakdown Structure (WBS)

Document Version: 1.0

Companion document to SRS.pdf – Folder: 4_SRS_and_WBS

Work Breakdown Structure

The project is decomposed into five phases. Each phase is broken into tasks, and each task into concrete sub-tasks / work packages that can be assigned, estimated, and tracked.

1. Requirements Engineering

1.1 Identify stakeholders

- Confirm actors: Student, Faculty Evaluator, System Administrator

1.2 Gather requirements

- Collect functional needs from stakeholder interaction
- Collect constraints and quality expectations

1.3 Define Functional Requirements (FR)

- Draft FR-001 to FR-005
- Assign priority and acceptance criteria to each FR

1.4 Define Non-Functional Requirements (NFR)

- Draft NFR-001 (performance/capacity)
- Draft NFR-002 (availability/data integrity)

1.5 Prepare Requirements Traceability Matrix (RTM)

- Map each FR/NFR to its source use case and test case ID

2. System Design

2.1 Architecture design

- Define submission intake, test-execution, scoring, and reporting components

2.2 UML use-case diagram

- Model actors, use cases, and system boundary
- Add include/extend relationships

2.3 Use-case flow

- Document main success scenario
- Document alternate / exception flows

2.4 Component / sequence design

- Draft component diagram for module interactions
- Draft sequence diagrams for key flows (submit, evaluate, review)

3. Development

3.1 Submission module

- ZIP upload endpoint and validation
- Submission ID generation and status tracking

3.2 Test execution

- Isolated execution environment / sandbox runner
- Syntax checks and unit test suite execution

3.3 Rubric evaluation

- Per-criterion scoring logic
- Faculty override with justification and audit trail

3.4 Report generation

- Score report (PDF) generation and export
- Peer-review feedback integration into the report

4. Testing

4.1 Test planning

- Define test strategy and environment
- Derive test cases from the RTM (TC-01 onward)

4.2 Functional testing

- Verify FR-001 to FR-005 against acceptance criteria

4.3 Integration testing

- End-to-end: submission → execution → scoring → report

4.4 Bug fixing and retesting

- Triage defects, fix, and retest
- Regression pass before release

5. Deployment & Documentation

5.1 Deployment

- Provision execution infrastructure
- Deploy and configure for the submission window load profile

5.2 Documentation

- Finalize SRS, WBS, RTM, and diagrams

- Prepare README and setup instructions

5.3 Final demonstration

- Prepare demo script and sample submissions
 - Record / deliver final walkthrough
-

Phase Summary

High-level view of the same breakdown, useful for scheduling and effort estimation. Fill in Owner and Estimated Effort columns per your team's sprint plan.

Phase	Title	# Tasks	Owner	Estimated Effort
1	Requirements Engineering	5	Owner	Est. hrs
2	System Design	4	Owner	Est. hrs
3	Development	4	Owner	Est. hrs
4	Testing	4	Owner	Est. hrs
5	Deployment & Documentation	3	Owner	Est. hrs